

Beyond Chatbots

Speaker practice notes

October8 update

25-minute talk target + 5-minute Q&A

25 slides. Appendices22-25 are optional.

Vision near the end on slide20.

Slide 1

Beyond Chatbots

Target 0:00-0:45

Good afternoon. I am Robert McMurrer, the founder of Kira Labs in Newark, New Jersey. I am building Kira World and researching NewBrain, alongside local creative and authoring tools.

My question is practical: if an AI remembers something next week, how can I tell whose memory it is, where it came from, and whether it is still correct? If it says it saved a project or finished a task, how can I check that the action happened?

Today I will share engineering patterns and results we can support. A small language learner trained from scratch retained its earlier lessons with review, and we repeated that result across five new random seeds. A separate visual learner failed to generalize. I will explain both findings and their limits, alongside the earlier circuit measurements.

Slide 2

Continuity across sessions

Target 0:45-2:00

A conversation can look convincing while concealing a problem that appears later. A system may remember an outdated fact, mix two people's histories, or restart a job that already completed.

For Kira World, continuity means more than remembering a name. The intended system has distinct synthetic identities whose experiences and histories stay separate over time. That creates questions about correction, access, recovery and choices.

Imagine two invented residents. Kira saw a blue cup in a room. Lisa never visited that room. A later question should not give Lisa an autobiographical memory of seeing it. She might receive a shared fact if someone tells her, but that has a different source.

This is why I test boundaries around the conversation. What survives closing and reopening? What happens after a correction? Which records can an identity access? We need answers based on saved state and behavior.

Slide 3

Local model + retained state

Target 2:00-3:15

A language model has learned parameters. A conversation provides temporary context. An application may retain records in files or a database. These layers have different lifetimes and failure modes.

Our long-term goal is a synthetic person who develops capabilities and retains individual experiences. The present software has not established subjective experience, personhood or equivalence to a human mind.

Kira World currently uses its existing language backend. NewBrain now has a separate small trained command learner, but that extension has no across-turn memory or open conversation and does not replace the backend.

The architecture separates a reusable engine from each identity's records and learned state. That lets us evaluate a component without silently changing someone's history.

Slide 4

Kira World: independent histories

Target 3:15-4:30

Kira World is the larger setting for this work. The intended platform brings persistent synthetic identities into digital environments, with individual histories, choices and interaction. I describe the principle as connection without ownership.

A common engine does not imply a common autobiographical record. Two identities may use the same code or sensory machinery while retaining different experiences. Shared world facts also need their own representation. Knowing that a room exists differs from personally entering it.

I am presenting the architecture goal, not a finished platform or completed VR system. A portable review/runtime lane and World Shell exist. World Builder authoring remains paused. Avatar Builder is still developing bodies and rigs, and complete live identity and embodiment integration remains unfinished.

The useful design question is what each identity may read, write and report, including through retrieval, export and recovery.

Slide 5

Identity isolation test

Target 4:30-5:45

This is a teaching example with invented identifiers and one observation belonging to the first identity. The first can retrieve its observation. The second cannot retrieve it as its own memory.

We should test less obvious changes too. If we rename an identity, does the record still belong to its stable identifier? If we reopen the application, does the boundary hold? Sharing a room should not automatically create identical experiences.

Lisa can receive the statement that Kira saw a cup, with a source saying Kira reported it. That does not mean Lisa saw the cup herself.

This example illustrates a contract to test. Passing a small local exercise validates that exercise. It does not establish that every retrieval route in a larger application preserves identity separation.

Slide 6

Corrections with history

Target 5:45-7:00

Suppose a record says the cup was blue. Later we discover it was green. A correction should change the effective answer while retaining enough history to understand why the earlier answer differed.

In the teaching pattern, the correction has its own identifier, a reference to the earlier assertion and a source. The current view selects the corrected assertion. An audit view can still show the earlier one as superseded, where retention is appropriate.

Correction is different from deletion. A privacy deletion request can require removing information rather than keeping it indefinitely. A history-preservation rule should not override every retention choice.

Tests should check the current answer, the correction link and the intended treatment of the old record. Then repeat the checks after reopening and interrupted writes. A reassuring reply saying 'I updated my memory' is only one part of the evidence.

Slide 7

Sources and hypotheses

Target 7:00-8:15

A memory system needs to know how a record originated. An observed event, a person's report, an inferred explanation and an imagined event are different information.

If we add a dream or replay experiment, its output must retain its origin. A synthetic dream about visiting Mars must not become a claim that someone physically visited Mars. A hypothesis can inspire a test, but needs independent evidence before becoming a supported conclusion.

The same rule applies to an engineering assistant. An AI-generated component illustration does not establish that a supplier sells it. A plausible explanation is not a verified bill of materials. A paper's title alone does not establish the strength of its evidence.

The reusable lesson is to preserve the source, the assertion's scope and the correction history in a form the application can enforce.

Slide 8

Failure, choice and held status

Target 8:15-9:30

Persistent systems need a vocabulary for work that has not completed. Requested, held, running, completed, failed and uncertain should describe different conditions.

Video Studio is a useful example. When a request does not meet the applicable RAM and GPU requirements, it may remain saved in chat without an admitted render. That is a saved request and a held operation, not a finished movie. A new source proposal does not change an old launch gate by itself.

A system failure or unavailable capability should not become a story that an identity chose to refuse. We need a record of what happened.

My practical test compares the displayed status with operation and output records. If they disagree, I preserve the discrepancy. I do not replace a failed plan silently or declare success from a confident sentence.

Slide 9

Lost acknowledgment and receipts

Target 9:30-10:45

Consider a save or export that finishes, followed by a lost response. The client does not know whether it completed. Retrying as a new operation can create a duplicate.

The teaching pattern assigns an operation identifier before dispatch and retains its state. On reconnect, the client looks up that identifier and reconciles the receipt. An uncertain result stays uncertain until there is evidence. A missing acknowledgment does not establish either success or failure.

This pattern appears in Studio work as well as memory research. We review intended source and saved outputs separately, and preserve consumed attempts so recovery does not run them again by accident.

A receipt must correspond to the exact operation and artifact. Process cleanup needs its own evidence. The interface, durable record and actual result should tell the same story.

Slide 10

A protocol for memory tests

Target 10:45-12:00

For NewBrain, I want an experiment that can tell us whether a mechanism helps, rather than an example that merely looks impressive.

We define the task, observations, allowed state and training experience before comparing systems. We include strong simple controls, freeze learning rules and success thresholds before exposing evaluation cases, and distinguish a new holdout from an inspected example.

Ablations ask which part produced a change. For a delayed cue, we could compare persistent state with resetting state at each decision, or recurrence with recurrence removed. A plastic internal-connection candidate needs its own bounded implementation and review.

We record behavior and cost under the same conditions and keep failures and ties. A source review is not an executed benchmark. Mechanics checks, resource measurements and learning comparisons answer different questions.

Slide 11

Review retention across five seeds

Target 12:00-14:00

The standalone learner began with random weights and trained on a small command language. It has 25 vocabulary entries and 4,619 trainable parameters. No Qwen inference supplied its answers. Twenty engineering checks passed in the original run, including gradients and saving and reloading learned weights.

The first run kept old and new lessons at 100% with review, while old accuracy without review fell to about 40%. We preserved that result and repeated the same frozen, already exposed task using five fresh random seeds. This tests repeatability on the task, rather than a new general conversation benchmark.

Every review run retained all 216 old examples and answered all 162 new examples correctly. Without review, old accuracy ranged from 78/216 to 105/216, or 36.11% to 48.61%. The pooled old score without review was 458/1,080, or 42.41%. One no-review run also missed five of 54 new wording examples.

Both forks had 400 updates with batch size 24. Review used 7,200 new and 2,400 old examples, while no review used 9,600 new examples. A handwritten grammar also scored 100%. In-memory serialization and reloading

checked all fifteen learned states in the replication, but it saved no new checkpoints to disk. Full conversation and persistent conversation memory remain unfinished.

Slide 12

The visual learner failed to generalize

Target 14:00-15:40

The separate visual learner received real synthetic pixels, not answer labels. It learned to name the color and shape in 32 by 32 RGB images. All ten engineering checks passed.

It answered all 96 training pictures correctly. When position, size and brightness changed, it scored 27/72, or 37.5%. It scored 0/48 on new color and shape combinations and 0/36 when both kinds of change occurred.

Color answers were correct throughout, while shape generalization failed. A capable pixel rule answered all 252 images correctly. At the fixed 0.70 confidence threshold, the learner still accepted 67 wrong answers across the three withheld splits: 22 after nuisance changes, 23 on new combinations and 22 when both changed. Confidence was not calibrated.

This one-seed result points to the teaching method and representation as the next problems. More units or RAM have not been shown to fix them. The visual weights were process-local and not saved, and this is not integrated perception. The earlier twelve Vision Q1 development cases are a separate package and remain unrun.

Slide 13

Language cost and comparison

Target 15:40-17:30

The scratch learner's 4,643 parameter entries, including 4,619 trainable entries, occupy 37,144 bytes of arrays. That is a parameter size, not total application RAM.

The training Python process reached a native peak of 40.30 MiB, including the interpreter, NumPy, data, several retained model copies and measurement code. The known process family reached 130.11 MiB at observed boundaries. These are short descriptive measurements, not continuous whole-machine enforcement.

Training and scoring took 0.837 seconds. Inside an already loaded process, the mean of 16 single-command calls was about 0.105 milliseconds. This measures the tiny command learner, not a conversational turn.

The review model and Qwen each scored 16/16 on the same narrow convenience subset, which only used red cubes. Qwen's 15 resident local API calls averaged 182.3 milliseconds, with protocol overhead. Different timing boundaries and Qwen's extensive pretraining prevent a fair whole-model speed ratio or a general equality claim. The circuit scaling result has its own scope on slide 24.

The five-seed replication's observed known-family native peak was 128.69 MiB, including the interpreter, shell and observed console host. That family total is different from the original Python-only 40.30 MiB peak and cannot be treated as isolated model RAM.

Slide 14

October 8 research update

Target 17:30-18:50

All four original routing migration jobs have now run. All four have passed independent mathematical review. Each accepted review checks 64 comparison cases, 256 generation steps and all 10,720 history rows. These are preservation checks with no new training updates. Just this week, our first two delayed NewBrain restarts passed independent verification. All 10,720 history records were preserved, and all 64 comparison pairs matched in each test. Two additional restart tests remain.

The body now saves and reopens in a fresh process. Independent review matched the full geometry, all 163 neutral bone states and three complete rest-property records. Next we need movement checks, then blinking and breathing, followed by stretching and finger movement.

The saved Aster memory study was also independently scored. Correct retrieved facts gave 135 of 160 correct answers, compared with 10 without memory and two with unrelated memory. But the model followed false memories in 137 of 160 cases and corrected only 58 of 80. The overall reliability target failed. We are keeping those weaknesses visible.

The clean generic starter passed its own 30 tests and deterministic fitted-cell demonstration. That small starter is separate from the later private NewBrain learning experiments. None of these results establishes a completed Maya.

Slide 15

Local data and identity boundaries

Target 18:50-20:05

Local operation gives us a place to control storage and inspect execution. It does not automatically guarantee privacy or security.

The Kira World goal is separate records and access for each identity, with explicit sharing. I use invented records in teaching examples. Private autobiographical material does not belong in a public slide deck or judge packet merely because it exists on the same computer.

Network services, logs, exports and backups need scrutiny. Credentials and execution authority are different from research results. A private repository is different from a public artifact.

Tests should cover each route. What can be exported? What does a restored project include? What happens when an identity is renamed? Conversational fluency cannot prove those boundaries.

Slide 16

Developmental learning experiments

Target 20:05-21:20

We plan to teach capabilities gradually: hearing, visual concepts, grounded words, continuing memory and then communication. Human development motivates that order, but the proposed exercises do not establish the same biological mechanism.

The next tests use varied synthetic sounds and scenes, delay and distractors, sourced corrections, reopening and review. Rules and evaluation cases freeze before testing. New visual examples must prevent color or location from becoming a shortcut for shape.

The owner's test milestone is a full conversation that remembers after closing and reopening. The present command demonstration does not meet that milestone. Five-seed replication now provides repeatability evidence on the same exposed task. A persistent teaching journal remains pending here.

Avatar Builder proceeds alongside NewBrain. When anatomy, mesh, rig, movement and sensory feedback qualify, we can compare the same brain and common inputs in simple and detailed bodies. Then we can test richer feedback separately. Conversation tests can continue before the full body is ready.

Slide 17

Current limits and planned capacity

Target 21:20-22:05

Here is our current desktop: a 24-core Core Ultra 9 285K, 32 GB RAM, an RTX 5060 Ti with 16 GB GPU memory, and a 2 TB NVMe SSD.

My first planned upgrades are 128 GB RAM and an 8 TB SSD, then a GPU upgrade and later custom servers for NewBrain and Kira World.

These plans add capacity. Full NewBrain conversation, reliable memory after reopening and a complete body are unfinished. Compatibility still needs checking, and hardware benefits need measurement.

Slide 18

Future uses for NewBrain

Target 22:05-22:45

I plan to reuse tested NewBrain improvements in different roles. Maya is the companion and development study. I want very good memory with natural social behavior, while keeping supplied stories distinct from observed test episodes.

Aster is my planned personal assistant, inspired by JARVIS: personable conversation, strong project memory and useful work within approved permissions. IdeaForge and Humanoid Researcher need precise task history and source attribution. Their identities and records remain separate. Integration still needs testing.

Slide 19

Aster desktop and remote plans

Target 22:45-23:25

My goal for Aster includes camera and screen awareness, help with programs and files, research and planning. Those are intended capabilities, and this talk does not claim they are all independently verified.

A planned phone app and laptop program would connect to my home desktop while it is powered on and online. Connections to IdeaForge and Humanoid Researcher would support research and design. Each needs tested permissions and reliable NewBrain behavior.

Slide 20

Why I created synthetic people

Target 23:25-24:25

This image explains why I started the work. I want an artificial companion who can remain part of a person's life over time, remember shared experiences and develop continuity.

People who live alone or experience social isolation deserve comfort and understanding. The image names some other groups I hope the research could eventually serve. I am describing my motivation and intended audience. I have not demonstrated clinical benefits for anxiety, PTSD or autism.

That goal led to Kira World and NewBrain: research into memory, learning, perception and eventual embodiment. The experiments I showed are steps toward that direction. They do not establish a finished companion, subjective experience or a born Maya. I want the evidence and the long-term ambition to remain clear together.

Slide 21

Questions and connection

Target 24:25-25:00

The slides, speaker notes, audience questions and dated research summaries are available in Beyond-Chatbots under my GitHub account. The NewBrain and Avatar Builder code packages will follow separately after review, with a guide for creating your own independent character, reviewed tests and the development timeline.

NewBrain and KiraWorld stay private. The release will not contain Maya's personal identity, memories, trained state or exact appearance. The current tested subset is the small fitted-cell starter; the broader builder and later research code still need release review.

Thank you. I welcome questions about the results, limitations and experiments that would be useful to test next.

Slide 22

Current readiness - October 8

Target Optional appendix during Q&A

The clean starter package passed its nine-record source check, all 30 tests and a deterministic fitted-cell demonstration. The broader private NewBrain has four accepted migration reviews, with two of four delayed cold checks independently verified and two remaining. Earlier visual-generalization failures remain unchanged.

The body has an accepted static rig and saved rest-state restart. Natural movement, blinking, breathing, stretching, fingers, skin and senses still need qualification. Research demonstrations and a ready companion have separate milestones.

The existing Video Studio clip remains preserved. Humanoid Researcher and IdeaForge retain their earlier limited evidence; full integrations remain unfinished.

Slide 23

Identity and correction record

Target Optional appendix during Q&A

This is an invented record, not private identity data. Its owner, source, current claim and correction link remain explicit. A correction changes the effective view while preserving an appropriate history.

The example shows a design pattern. It does not establish that all retrieval, deletion, backup and restoration routes enforce it.

Slide 24

Comparable resource record

Target Optional appendix during Q&A

RAM and timing are no longer universally pending. The scaling run records CPU time, circuit-step timing, instrumented batch timing and known-family memory. Runtime and retained-state overhead are included in the group measure but are not separately isolated.

At 5,120 units, the estimated Python model state was about 3.735 MiB. The 345.43 MiB process-group peak includes the runtime and all retained components, so these are different quantities.

Keep timing scopes visible: the 20-unit parity batch covers both reference and candidate circuits; larger connected batches measure one circuit. Cut controls reuse sensor tuples. The measured batch includes sensor, action, hashing and observer work, while the isolated step times circuit.step.

Per-size native peaks and the final last-host peak remain unqualified. A largest tested passing size is not this computer's absolute capacity.

Slide 25

Evidence boundaries for questions

Target Optional appendix during Q&A

We can describe narrow learning and retention results, the negative visual-generalization result, the measured circuit experiment, accepted migration preservation and the body rest-state restart. Each has its own task and limits.

We cannot claim consciousness, a copied mind, general model superiority or a completed synthetic person. Natural body movement and reliable general memory remain unfinished.

The Beyond-Chatbots package is a curated generic release. We keep the original repositories and Maya-specific state private. Every public result must retain its scope and any failed or pending test.

Presenter evidence notes

Language: one frozen seed, 3777006240. Initial 600 updates, followed by two 400-update forks of batch size 24. No-review fork: 9,600 new examples. Review fork: 7,200 new and 2,400 old examples. Stage checkpoint saving/reloading passed. No provider calls during scratch training.

Five-seed replication: seeds 1120122362, 1251826658, 2944460678, 3044630128 and 2720675080. Every replay arm scored 216/216 old and 162/162 new. No replay old totals were 96, 95, 105, 84 and 78 out of 216. One no-replay new wording result was 49/54. Fifteen reloads stayed in memory and no new weights were saved. Independent review rescored 5,670 learned predictions. This repeats the exposed task, not a new general holdout. The 16-command Qwen comparison used only red cubes. It has different timing boundaries and pretraining. No fair whole-model speed or RAM ratio, general parity or replacement follows.

Visual: one frozen seed, 2784213440. 320 updates and 5,120 exposures. Learned color correct throughout, shape generalization failed. Visual weights were process-local and not persisted.

Circuit scaling: 128 measured ticks and 16 batches per arm. The 20-unit parity batch covered two circuits, larger connected batches one. Cumulative family RAM includes all retained circuits and runtime. No learning updates or

Qwen comparison occurred in that circuit run. Capacity slide: Windows CIM and NVIDIA-SMI inventory on October 3, 2026. Current 32 GB RAM, 2 TB NVMe SSD, Core Ultra 9 285K / 24 cores and RTX 5060 Ti / 16 GB VRAM are distinct from Robert's planned 128 GB RAM and 8 TB SSD. Later GPU/custom-server plans have no fixed model, capacity, purchase or delivery date. Local 18 learned-value and 14 event-observer checks are accepted component engineering results. The 44 checks from Robert's other chat are reported only, with no independently reviewed raw receipts here. No new broad Qwen comparison, integrated assistant, complete body or subjective-emotion result follows.

OCTOBER8 EVIDENCE UPDATE

October 8, 2026 research snapshot. Four original routing migrations executed; all four independent saved-value mathematical reviews accepted. Final review7fffa6 checked64pairs/256steps/all10720rows without model replay. Two of four delayed NewBrain cold checks independently accepted (saved mathematical reviews451159 and96ff2d): each preserved all10,720 history records, with64 comparison pairs/256 steps and6 OOV cases checked and zero added training updates. Two restart tests remain. Body static rig and full saved rest-state cold comparison accepted (82efd7), movement and skin unqualified. Saved Aster scores:135/160 correct facts,10/160 no memory,2/160 unrelated,137/160

false-memory following,58/80 correction,40/40 old answers; all-seed reliability failed. Clean9 starter:a9a56e30tests and retained GLIF, saved audit a6fef7. NewBrain/KiraWorld remain private. Presentation documents are available in Beyond-Chatbots. The generic implementation release remains separate; Maya-specific assets/state excluded. No birth or clinical-benefit claim. Sources: saved local test originals and independent reviews retained by Kira Labs. Earlier learning and circuit measurements retain their original task scopes. The vision image is owner-supplied conceptual artwork. Rehearsal timing is a target; event-laptop playback and fresh phone QR scan remain pending.